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Velodrome 300 S Smart Bike

SERVICE MANUAL

FF-BB-VELODROME-300-S

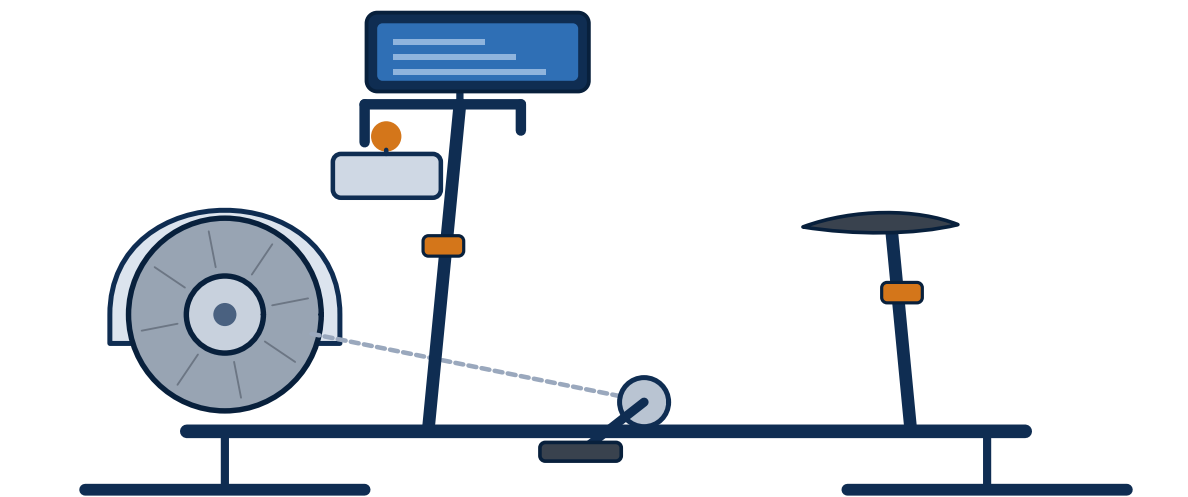


Illustration shows the machine in the running position. Specifications may vary by market.

This manual is intended for use by the equipment owner and by FitForge authorised service technicians.
Read Section 2 in full before carrying out any procedure.

Document FF-SM-ME-300-S · Revision C · Model year 2026 · Smart Bike

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SECTION 1 — MODEL IDENTIFICATION

Confirm the model before ordering parts or following any procedure in this manual. Procedures and part numbers differ between models in the same range, and fitting the wrong part will not resolve the fault.

SERIAL PLATE LOCATION

Underside of the frame, on the front stabiliser bar.
Fold the deck up and look down the left-hand rail.

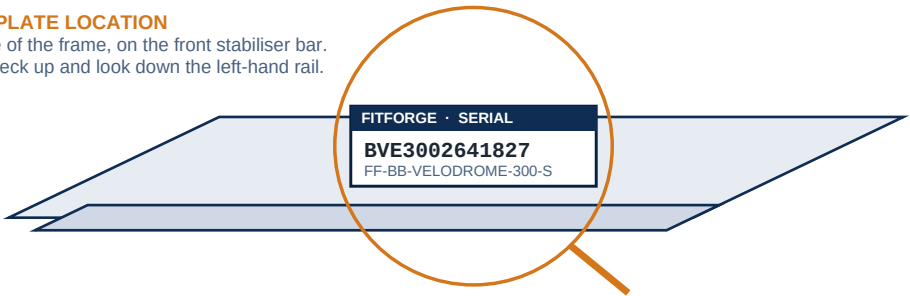


Figure 1 — Serial plate location

The plate is fixed on the frame just behind the crank, on the non-drive side.

Serial numbers for this model begin with **BVE300**, followed by a two-digit year and a five-digit sequence.

If the plate is missing or unreadable, the following characteristics distinguish this model from others in the range:

Characteristic	This model
Pedals	flat
Console	no screen
Resistance	magnetic

SECTION 2 — SAFETY

Read this section completely before operating or servicing the equipment. The procedures in this manual assume the machine is unplugged unless a step explicitly states otherwise.

⚠ DANGER — FIXED-GEAR FLYWHEEL

The flywheel is directly coupled to the pedals and does not freewheel. It carries enough momentum to keep the cranks turning hard after you stop pedalling.

Bring the pedals to a complete stop using the emergency brake lever before dismounting or reaching near the flywheel.

Never remove the flywheel shroud, and never operate the machine with it removed or damaged.

⚠ CAUTION — CLAMPS AND PINCH POINTS

Check both the seat post and handlebar post clamps before every use. A clamp that slips under load will drop the rider onto the frame.

Keep children and pets away from the equipment at all times. The gap between the crank and the frame is a pinch point even when the machine is unpowered.

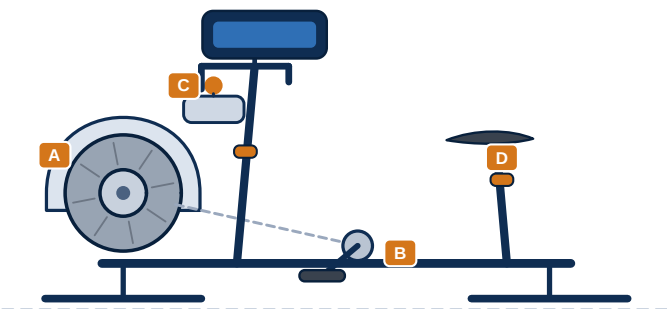
Position the machine on a level surface with at least 0.6 m of clear space on every side.

General requirements

- The flywheel is fixed-gear and does not freewheel. It will keep the pedals turning after you stop. Never dismount before the pedals stop.
- Never run the machine with the flywheel shroud removed.
- Check the seat post and handlebar post clamps before every use. Never raise either beyond its minimum-insertion mark.
- Keep children and pets at least two metres from the machine while it is in use.
- Do not exceed the maximum user weight given in Section 3.
- Use only the supplied 12 V adaptor. There is no mains voltage inside the frame, and no user-serviceable part in the adaptor.

Safety decal locations

Replace any decal that is missing, torn or illegible before returning the machine to service. Order replacements by position reference.



DECAL	POSITION
A	Flywheel pinch warning — shroud
B	Max user weight — frame, near crank
C	Read the manual — handlebar post
D	Minimum insertion — seat post

Figure 2 — Safety decal locations

SECTION 3 — SPECIFICATIONS

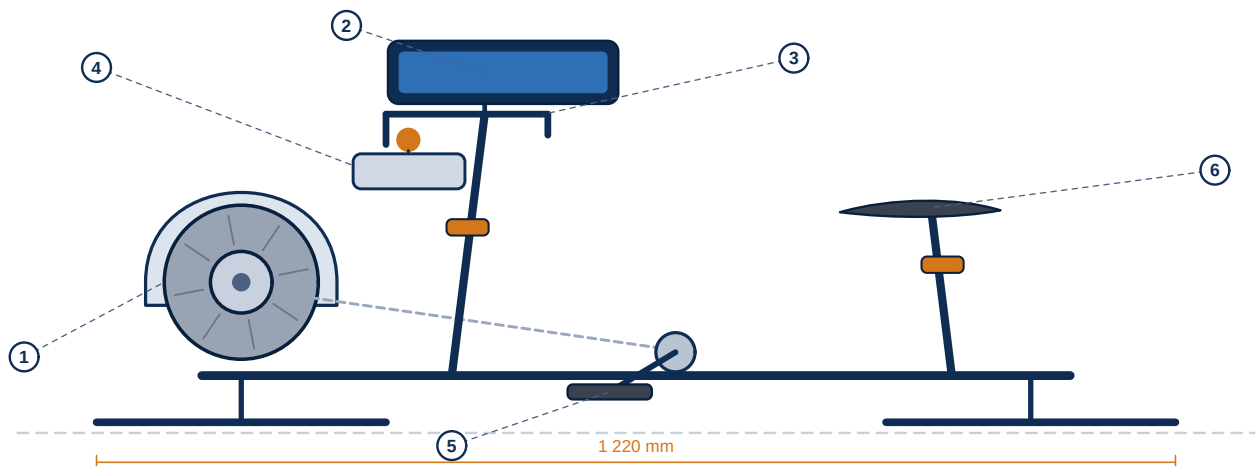


Figure 3 — Principal assemblies

#	Assembly	Service note
1	Flywheel and shroud	Do not run without the shroud fitted. Technician only.
2	Console assembly	Customer replaceable. Unplug the supply before removal.
3	Handlebar post and bars	Check the clamp before every use.
4	Resistance actuator	Calibrate from Settings > Device > Calibrate.
5	Crank arm and pedals	Re-torque at 30 days, then every 6 months.
6	Seat post assembly	Never raise beyond the minimum-insertion mark.

Technical data

Parameter	Value
Flywheel	18 kg, perimeter weighted
Drive	Toothed belt. No chain maintenance.
Resistance	magnetic
Resistance levels	100, micro-adjustable
Console	no screen
Pedals	flat
Q-factor	168 mm
Adjustment	Saddle and bars, 4-way each
Maximum user weight	150 kg
Unit weight	58 kg
Supply	External 12 V DC adaptor. No mains voltage inside the frame.

Torque specifications

Fastener	Size	Torque
Pedal into crank arm	9/16 in	35 Nm. Right hand thread; left is reversed
Crank arm to spindle	M8	40 Nm
Seat post clamp	M6	12 Nm. Never exceed

Fastener	Size	Torque
Handlebar post clamp	M6	12 Nm
Stabiliser to frame	M8	24 Nm
Console to post	M5	6 Nm

⚠ CAUTION - SEAT POST INSERTION

The seat post carries a minimum-insertion mark. Raising the post beyond it puts the whole rider load on the top of the frame tube rather than on the sleeve. It will crack the frame, and it voids coverage on both the frame and the post.

SECTION 4 — MAINTENANCE SCHEDULE

The majority of service calls are caused by missed maintenance. Following this schedule prevents most of the faults described in Section 5.

Interval	Task
Every use	Wipe sweat from the frame, bars and post. Sweat corrodes the clamps.
Weekly	Check the pedals are tight in the crank arms. Vacuum beneath the unit.
Monthly	Check seat and handlebar clamp torque. Inspect the adaptor lead.
Every 3 months	Check drive belt tension. Inspect for fraying along the tooth line.
Every 6 months	Re-torque all frame fasteners against the table in Section 3.
Every 12 months	Inspect flywheel bearings for play. Calibrate the resistance actuator.

Pedal and crank torque check

1. Unplug the adaptor and wait for the flywheel to stop completely.
2. Hold the crank arm and try to rock the pedal by hand. Any movement means the pedal is loose.
3. The left pedal has a REVERSED thread. Turn it anti-clockwise to tighten and clockwise to remove.
4. Torque both pedals to 35 Nm.
5. Check the crank arm on the spindle the same way, and torque to 40 Nm if there is any movement.

SECTION 5 — TROUBLESHOOTING

Work through the checks for a symptom in the order given. They are ordered cheapest and safest first, and a later check often assumes an earlier one has been completed. Do not skip ahead.

Symptom: screen is blank or stuck on the logo

Also described as: blank screen, wont boot, stuck on startup, display dead.

1. Confirm the power brick LED is lit and the barrel connector is fully seated at the frame.
2. Hold the power button for 20 seconds to force a hard reset.
3. If the screen shows the logo then goes dark, the display is receiving power but failing to boot — try the update-recovery combination in the manual.
4. Reseat the display ribbon cable behind the handlebar post cover.

If the checks above do not resolve the symptom, the following parts are the likely cause, most likely first: display, psu.

Symptom: resistance does not change when adjusted

Also described as: no resistance, resistance stuck, always easy, always hard.

1. Confirm the resistance value on screen actually changes when you adjust it — if the number is frozen, this is a display or software fault, not mechanical.
2. Run resistance calibration from Settings > Device > Calibrate.
3. Listen for the actuator motor during calibration; a healthy actuator makes a clear whirring sweep.
4. If there is no actuator sound at all, check the actuator connector at the mainboard.

If the checks above do not resolve the symptom, the following parts are the likely cause, most likely first: resistance motor.

Symptom: clicking or knocking noise when pedalling

Also described as: clicking, knocking, creaking, tick each rotation.

1. Pedal slowly and note whether the noise happens once per crank revolution.
2. Check both pedals are torqued to 35 Nm — the left pedal is a LEFT-HAND thread.
3. If tightening the pedals does not fix it, remove them and apply a thin film of grease to the threads.
4. Check the crank arm bolts at the bottom bracket.

If the checks above do not resolve the symptom, the following parts are the likely cause, most likely first: pedal set, crank arm.

Symptom: power or cadence readings are zero or erratic

Also described as: no watts, no cadence, power drops out, erratic readings.

1. Confirm the ride screen is showing a live cadence figure while you pedal.
2. Perform the zero-offset calibration with the cranks stationary and no weight on the pedals.
3. Check the sensor cable at the bottom bracket for a loose or damaged connector.
4. If readings drop out only under high load, the sensor mount has flexed and needs re-shimming.

If the checks above do not resolve the symptom, the following parts are the likely cause, most likely first: power meter.

Symptom: seat slips down during a ride

Also described as: seat sinks, saddle drops, post slips.

1. Confirm the seat post clamp lever is closed to a firm, not merely snug, position.
2. Wipe the seat post and the inside of the frame collar — grease on the post is the most common cause.
3. Check the clamp adjustment nut; the lever should require real force in the last 20 degrees of travel.
4. Inspect the post for scoring, which prevents the clamp gripping.

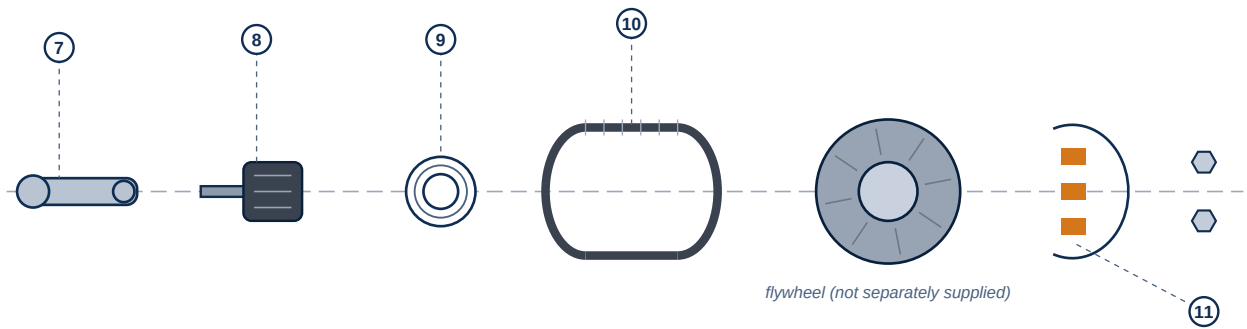
If the checks above do not resolve the symptom, the following parts are the likely cause, most likely first: seat post.

SECTION 6 — ERROR CODES

Codes are shown on the console. Record the code before power cycling — it is not retained across a restart.

Code	Meaning	First actions
B1	Resistance actuator out of range. The actuator did not reach the commanded position.	Run resistance calibration from Settings > Device > Calibrate. If it fails twice the actuator has failed.
B2	Power meter signal invalid. Cadence or torque readings are outside plausible bounds.	Check the sensor cable at the bottom bracket is seated. Re-run the zero-offset calibration with the cranks stationary.
B3	Display panel not detected. The mainboard cannot enumerate the display.	Power cycle by unplugging for 60 seconds. Reseat the display ribbon cable behind the handlebar post cover.
B6	Firmware update failed. The last over-the-air update did not complete.	Reconnect to Wi-Fi and retry from Settings > System > Update. The bike will fall back to the previous firmware automatically.

SECTION 7 — PARTS LIST



Shown exploded along the drive axis. Reassemble in reverse order.

Figure 4 — Drive assembly, exploded view

#	Part number	Description	Cust. fit
7	FF-BB-VELODROME-300-S-CRANK-ARM	Crank arm, 170 mm	Yes
8	FF-BB-VELODROME-300-S-PEDAL-SET	Pedal set, dual SPD and cage	Yes
9	FF-BB-VELODROME-300-S-FLYWHEEL-BEARING	Bottom bracket and flywheel bearing set	No
10	FF-BB-VELODROME-300-S-BELT-DRIVE	Toothed drive belt	Yes
11	FF-BB-VELODROME-300-S-RESISTANCE-MOTOR	Magnet carrier and resistance actuator	No

Full parts list

Quote the complete part number when ordering. Parts marked technician only must not be fitted by the customer; doing so ends the remaining coverage on the machine.

Part number	Description	Class	Cust. fit
FF-BB-VELODROME-300-S-DISPLAY	Touchscreen Display	Electronics	Yes
FF-BB-VELODROME-300-S-RESISTANCE-MOTOR	Resistance Actuator	Electronics	Yes
FF-BB-VELODROME-300-S-CRANK-ARM	Crank Arm	Mechanical	Yes
FF-BB-VELODROME-300-S-PEDAL-SET	Pedal Set	Consumable	Yes
FF-BB-VELODROME-300-S-BELT-DRIVE	Drive Belt	Mechanical	Yes
FF-BB-VELODROME-300-S-FLYWHEEL-BEARING	Flywheel Bearing Set	Mechanical	Yes
FF-BB-VELODROME-300-S-POWER-METER	Power Meter Sensor	Electronics	Yes
FF-BB-VELODROME-300-S-SEAT-POST	Seat Post Assembly	Mechanical	Yes
FF-BB-VELODROME-300-S-HANDLEBAR-POST	Handlebar Post	Mechanical	Yes
FF-BB-VELODROME-300-S-PSU	Power Supply Unit	Electronics	Yes

SECTION 8 — WARRANTY

Component group	Coverage from date of purchase
Frame	120 months
Mechanical parts	36 months
Electronics	24 months
Labour	12 months
Wear and consumable items	90 days, manufacturing defects only

Coverage begins on the date of purchase recorded against the serial number. Commercial or institutional use voids parts and labour coverage under the residential warranty. Wear items — belts, decks, rollers, straps, pads and lubricant — are covered for manufacturing defects for 90 days from purchase and are not covered thereafter.

To make a claim, contact FitForge support with the serial number and the order number. Retain proof of purchase; coverage attaches to the purchase record rather than to the owner.